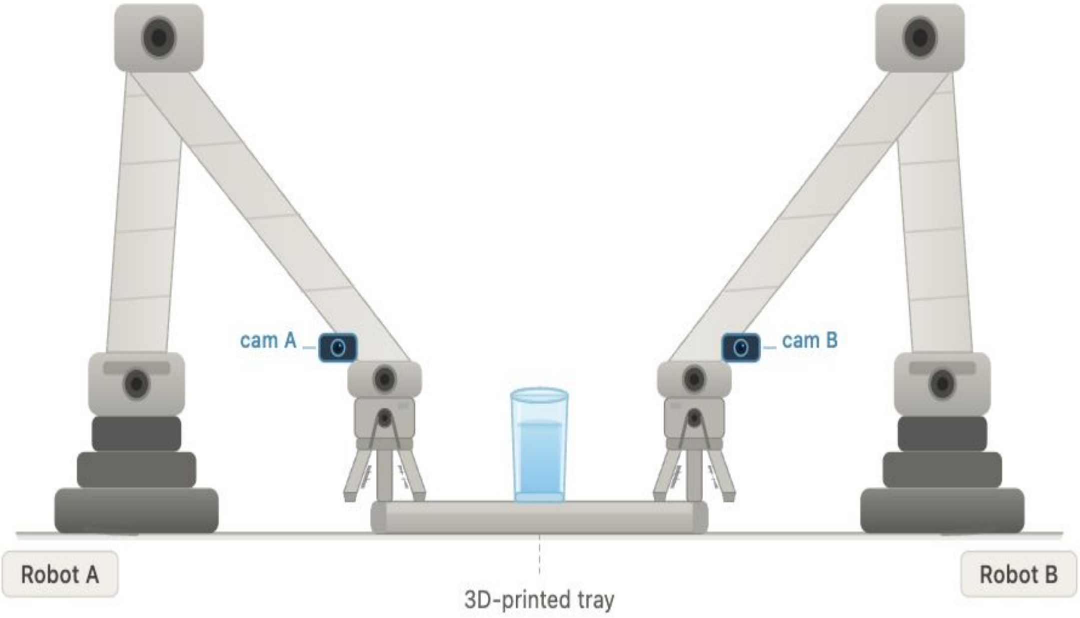
GNN for Autonomous Robot Arm Control

Jackson Crocker and David Herrera

- Two robot arms cooperatively carry a tray with a water glass using only wrist cameras (no communication)
- A GNN maps visual observations to arm states, with edges representing learned transitions
- Visual message passing enables implicit inter-arm coordination without explicit communication
- Trained on human demonstrations collected via UMI-style hand-held grippers



AI Visualization

Project Status

- All required materials arrived
- Both arms are printed, assembled, and functional - able to complete task with manual input
- Data collection and control pipeline (camera -> laptop -> servo motors) configured but not yet utilized

Semester Plan

- Design and implement GNN
  - Literature review for initial approaches
  - Huge amount of data collection
- Comparison to existing solutions
- Exploration of generalizability of algorithm

Metric for success:
Robots coordinate without explicit communication to lift and move tray without spilling glass of water.

Previous Work

- Jlidi, A., Benotsmane, R., Kovács, L., & Trohak, A. (2025).

  A novel graph neural network approach for inverse kinematics in robotic arms. In 26th International Carpathian Control Conference (ICCC). IEEE. DOI:10.1109/ICCC65605.2025.11022859
- d’Aquino Hilt, F., Kolf, J. N., Weiland, C., & Carvalho, J. (n.d.).

  Graph neural networks for robotic manipulation.

  Technical University of Darmstadt.